Option Pricing Report · AAPL

European call, strike 350.00, maturity 0.5000 years

345.37	26.7748	26.36%	76%	157/157
Spot (put-call parity)	Closed-form Black-Scholes	Implied volatility	ITM quotes inside bid/ask	Tests passed

Stale interest rates
The Treasury rates (FRED files in `data/raw/DGS`) end on 2026-05-04, 177 days before the valuation date. Download fresh DGS series from FRED. Until then r(T) is out of date; the implied carry absorbs most of the error in forwards, but discounting stays off by roughly the rate change times maturity.

Generated 2026-10-28 16:21 from Yahoo Finance market data (snapshot 2026-10-28 16:21:01). This report explains what the engine does, the inputs it used, the prices and Greeks it produced, how those prices compare with live market quotes, how the code is tested, and the known limitations to keep in mind when using the numbers.

How the engine works

Every price goes through the same pipeline. The CLI (`main.py`) and the FastAPI service (`/price`, `/market-data`, `/report`) are thin layers over it.

Step	Module	What it does
1. Market data	market_data	Fetches spot, Treasury yields (local FRED files), dividends and the full option chain.
2. Spot and carry	calibrators/implied_spot	Backs the spot and a carry curve q(T) out of put-call parity, so the inputs agree with the option quotes.
3. Implied volatility	calibrators/implied_volatility	Newton-Raphson inversion of Black-Scholes on liquid out-of-the-money quotes.
4. Vol surface	models/volatility_surface	Cubic spline smile per expiry in log-moneyness, linear in total variance across expiries.
5. Snapshot	market_data_snapshot	Freezes spot, curves and surface with a timestamp and source: the contract with the pricers.
6. Model and process	models, process	Black-Scholes with r(T), q(T), sigma(K, T) read for the option; geometric Brownian motion.
7. Pricers	pricers	Closed form, CRR binomial tree (European and American), Monte Carlo (exact GBM or Euler).
8. Greeks	greeks	Bump-and-reprice finite differences, identical for every pricer.

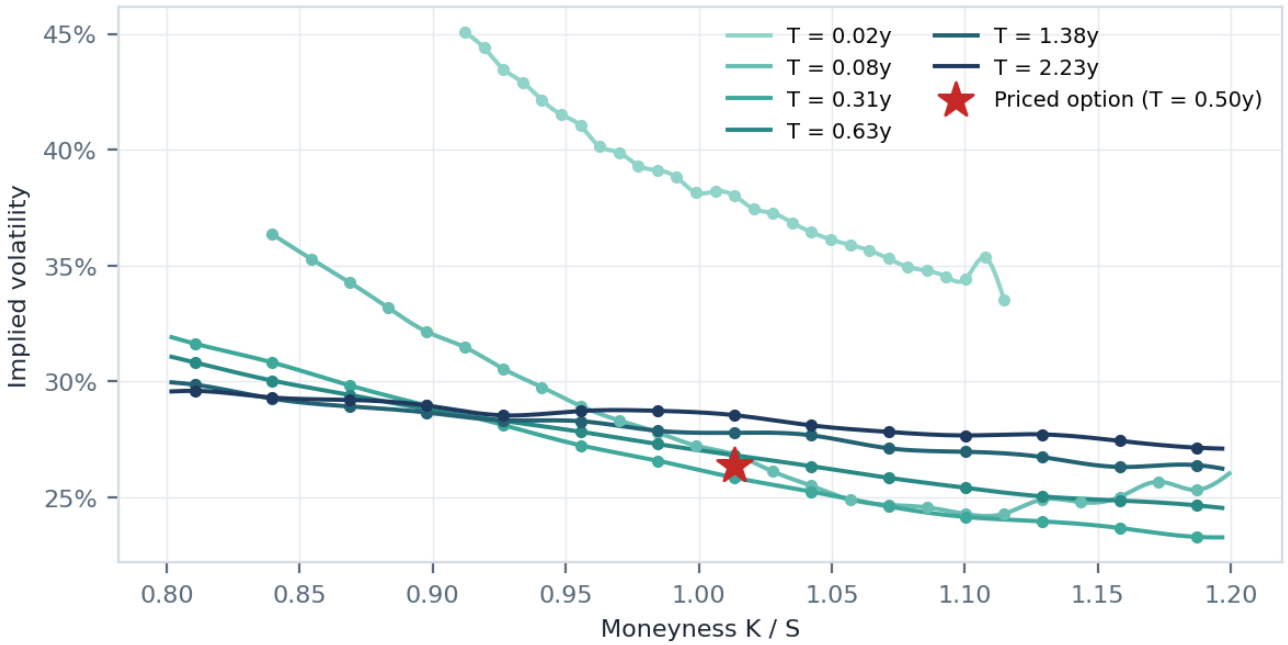
Market inputs

Ticker	AAPL
Spot used	345.3653 (put-call parity); quoted feed 345.1000, gap 0.077%
Risk-free rate r(T)	3.7600% (Treasury zero rate interpolated at T)
Carry / dividend yield q(T)	-0.2631% (put-call parity)
Implied volatility sigma(K, T)	26.3573% (read from the surface at the option's strike and maturity)
Surface expiries	18 smiles
Data source / time	Yahoo Finance, 2026-10-28 16:21:01

Volatility surface

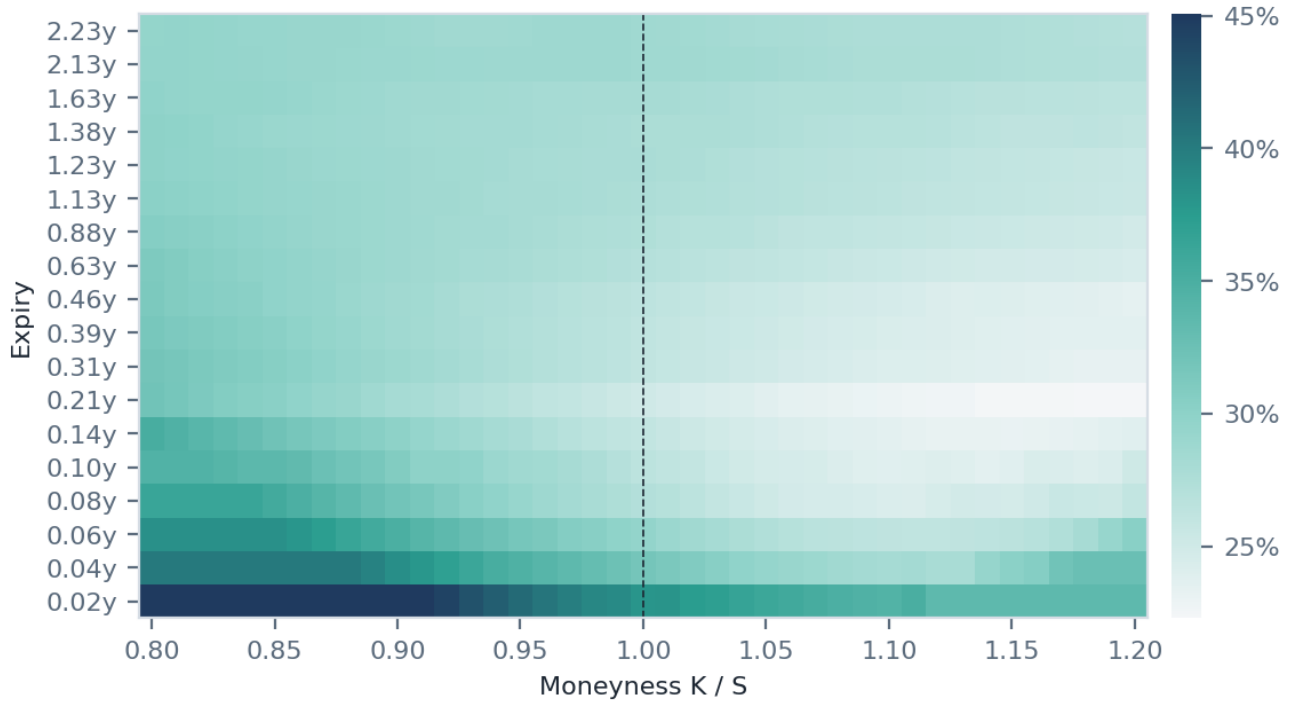
The implied volatility surface every option is priced from. Each expiry has its own smile, a natural cubic spline through the implied volatilities of liquid out-of-the-money quotes; between expiries the surface is linear in total variance.

Smiles



Dots are the calibration quotes, lines the fitted smiles, for up to six expiries spread across the surface. The star is the option priced in this report. Outside the quoted strikes the smile is extended flat.

Surface



Every quoted expiry; the dashed line is at the money. Darker means higher volatility.

Prices

Pricer	Price	vs closed form	Relative	Settings
Closed-form Black-Scholes	26.7748	0.0000	0.000%	Analytic formula
Binomial tree (CRR)	26.7741	-0.0007	-0.003%	1000 steps
Monte Carlo	26.7909	0.0162	0.060%	100,000 paths, 64 steps, exact scheme, variance reduction: antithetic, seed 42

The closed form is exact under Black-Scholes, so differences are the numerical error of the tree and of Monte Carlo.

Monte Carlo: standard error 0.0749, 95% confidence interval [26.6442, 26.9377] from 100,000 observations.

Greeks

Greeks are computed the same way for every pricer: bump one input, reprice, and take a central finite difference. Bump sizes come from `config.ini`. The volatility bump moves sigma of the process with the spot fixed, so these are *sticky-strike* Greeks. For Monte Carlo every bumped price reuses the same random numbers (common random numbers); otherwise the differences would be dominated by simulation noise.

Greek	Bump	Formula
Delta	spot ± 2.0% (h = 6.91)	(V(S+h) - V(S-h)) / 2h
Gamma	same spot bumps	(V(S+h) - 2V(S) + V(S-h)) / h²
Vega	sigma ± 0.50 vol points	(V(σ+d) - V(σ-d)) / 2d, per 1.00 of vol
Rho	r ± 1 bp	(V(r+d) - V(r-d)) / 2d, per 1.00 of rate
Theta	maturity ± 1 day	-(V(T+d) - V(T-d)) / 2d, per year

Results

Pricer	Delta	Gamma	Theta	Vega	Rho
Closed-form Black-Scholes	0.55204	0.00615	-32.16457	96.73818	81.98198

Pricer	Delta	Gamma	Theta	Vega	Rho
Binomial tree (CRR)	0.55248	0.00631	-32.12116	96.57479	81.97796
Monte Carlo	0.55182	0.00610	-32.00045	96.11308	81.99772
Analytic reference	0.55228	0.00615	-32.16447	96.73825	81.98198

Largest relative deviation from the analytic Greeks: Closed-form Black-Scholes 0.08%; Binomial tree (CRR) 2.53%; Monte Carlo 0.85%. Binomial gamma oscillates with the step count (odd / even nodes), which is normal for trees.

Monte Carlo accuracy

The same option is priced by Monte Carlo with each variance reduction technique and several path counts, all with the same seed, and compared with the closed-form price. **Antithetic** averages each path with its mirror (Z and -Z), so it simulates twice the paths. **Control variate** corrects the payoff with the discounted terminal stock price, whose expectation is known. The variance reduction factor is the variance without reduction divided by the variance with it, at the same number of paths: the number of plain paths each path is worth.

Technique	Paths	Price	Std. error	95% interval	Contains	Var. red.	Time
None	10,000	26.1660	0.4241	[25.335, 26.997]	yes	1.0x	16 ms
None	50,000	26.9182	0.1915	[26.543, 27.293]	yes	1.0x	89 ms
None	100,000	26.9672	0.1357	[26.701, 27.233]	yes	1.0x	166 ms
Antithetic	10,000	26.5791	0.2359	[26.117, 27.042]	yes	3.2x	19 ms
Antithetic	50,000	26.6609	0.1052	[26.455, 26.867]	yes	3.3x	134 ms
Antithetic	100,000	26.7909	0.0749	[26.644, 26.938]	yes	3.3x	273 ms
Control variate	10,000	26.6614	0.1916	[26.286, 27.037]	yes	4.9x	15 ms
Control variate	50,000	26.6656	0.0857	[26.498, 26.833]	yes	5.0x	81 ms
Control variate	100,000	26.7692	0.0608	[26.650, 26.888]	yes	5.0x	163 ms

Closed-form reference price: 26.7748. The standard error falls like 1 / sqrt(paths). Roughly one interval in twenty is expected to miss the closed form at 95% confidence, so an occasional miss is not a defect.

Model vs market

Every liquid listed option is repriced from the same snapshot and compared with its live bid and ask. Liquid means a real two-sided quote, a bid/ask spread under 10% of mid, strikes between 85% and 115% of spot and maturities from 30 days to 1 year. A price counts as a hit when it lies inside [bid - 0.01, ask + 0.01].

In-sample quotes are out-of-the-money: they are the quotes the volatility surface is built from, so this checks the calibration. **Out-of-sample** quotes are in-the-money: they never enter the surface, so this is the real test of the model and of the spot and carry calibration. Listed equity options are American, so in-the-money puts are priced with early exercise on the binomial tree.

Summary

Group	Quotes	Inside bid/ask	Mean abs. error	Median abs. rel. error
In-sample (OTM, calibration)	118	100.0%	0.000	0.00%
Out-of-sample (ITM)	131	76.3%	0.301	0.58%
All quotes	249	87.6%	0.158	0.02%

All quotes

Type	Strike	T (y)	Sample	Bid	Ask	Mid	Model	Error	Rel.	Hit
call	295.00	0.099	out	50.90	53.35	52.125	52.182	+0.057	+0.11%	yes
call	300.00	0.099	out	46.20	48.60	47.400	47.466	+0.066	+0.14%	yes
call	305.00	0.099	out	41.95	44.05	43.000	42.844	-0.156	-0.36%	yes
call	310.00	0.099	out	37.35	39.30	38.325	38.317	-0.008	-0.02%	yes
call	315.00	0.099	out	33.10	35.15	34.125	33.825	-0.300	-0.88%	yes
call	320.00	0.099	out	28.70	30.70	29.700	29.809	+0.109	+0.37%	yes
call	325.00	0.099	out	24.90	26.85	25.875	25.677	-0.198	-0.77%	yes
call	330.00	0.099	out	21.25	22.70	21.975	21.970	-0.005	-0.02%	yes
call	340.00	0.099	out	14.55	15.95	15.250	15.232	-0.018	-0.12%	yes
call	345.00	0.099	out	11.75	12.65	12.200	12.200	+0.000	+0.00%	yes
call	350.00	0.099	in	9.40	10.00	9.700	9.700	+0.000	+0.00%	yes
call	355.00	0.099	in	7.25	7.65	7.450	7.450	+0.000	+0.00%	yes
put	325.00	0.099	in	4.20	4.45	4.325	4.325	+0.000	+0.00%	yes
put	330.00	0.099	in	5.35	5.85	5.600	5.600	+0.000	+0.00%	yes
put	335.00	0.099	in	6.80	7.40	7.100	7.100	+0.000	+0.00%	yes
put	340.00	0.099	in	8.50	9.15	8.825	8.825	+0.000	+0.00%	yes
put	345.00	0.099	in	10.50	11.05	10.775	10.775	+0.000	+0.00%	yes
put	350.00	0.099	out, Am.	12.85	13.80	13.325	13.351	+0.026	+0.20%	yes
put	355.00	0.099	out, Am.	15.60	16.65	16.125	16.115	-0.010	-0.06%	yes
put	360.00	0.099	out, Am.	18.70	20.30	19.500	19.211	-0.289	-1.48%	yes
put	365.00	0.099	out, Am.	22.05	24.05	23.050	22.819	-0.231	-1.00%	yes
put	370.00	0.099	out, Am.	26.00	27.90	26.950	26.749	-0.201	-0.75%	yes
put	375.00	0.099	out, Am.	30.10	31.95	31.025	30.873	-0.152	-0.49%	yes
put	385.00	0.099	out, Am.	38.75	41.35	40.050	39.987	-0.063	-0.16%	yes
call	295.00	0.137	out	52.35	53.90	53.125	53.287	+0.162	+0.30%	yes
call	300.00	0.137	out	48.45	49.15	48.800	48.697	-0.103	-0.21%	yes
call	305.00	0.137	out	43.75	44.75	44.250	44.203	-0.047	-0.11%	yes

Type	Strike	T (y)	Sample	Bid	Ask	Mid	Model	Error	Rel.	Hit
call	310.00	0.137	out	39.75	40.15	39.950	39.848	-0.102	-0.26%	yes
call	315.00	0.137	out	35.55	36.25	35.900	35.548	-0.352	-0.98%	yes
call	320.00	0.137	out	31.50	31.80	31.650	31.474	-0.176	-0.56%	no
call	325.00	0.137	out	27.25	28.00	27.625	27.574	-0.051	-0.19%	yes
call	330.00	0.137	out	23.85	24.00	23.925	23.849	-0.076	-0.32%	yes
call	335.00	0.137	out	20.20	20.75	20.475	20.324	-0.151	-0.74%	yes
call	340.00	0.137	out	17.15	17.40	17.275	17.125	-0.150	-0.87%	no
call	345.00	0.137	out	14.20	14.45	14.325	14.325	+0.000	+0.00%	yes
call	350.00	0.137	in	11.60	11.80	11.700	11.700	+0.000	+0.00%	yes
call	355.00	0.137	in	9.30	9.45	9.375	9.375	+0.000	+0.00%	yes
call	360.00	0.137	in	7.30	7.50	7.400	7.400	+0.000	+0.00%	yes
call	365.00	0.137	in	5.60	5.80	5.700	5.700	+0.000	+0.00%	yes
call	370.00	0.137	in	4.30	4.45	4.375	4.375	+0.000	+0.00%	yes
call	375.00	0.137	in	3.25	3.35	3.300	3.300	+0.000	+0.00%	yes
call	380.00	0.137	in	2.41	2.49	2.450	2.450	+0.000	+0.00%	yes
put	295.00	0.137	in	1.43	1.50	1.465	1.465	+0.000	+0.00%	yes
put	300.00	0.137	in	1.82	1.88	1.850	1.850	+0.000	+0.00%	yes
put	305.00	0.137	in	2.32	2.34	2.330	2.330	+0.000	+0.00%	yes
put	310.00	0.137	in	2.91	2.99	2.950	2.950	+0.000	+0.00%	yes
put	315.00	0.137	in	3.60	3.65	3.625	3.625	+0.000	+0.00%	yes
put	320.00	0.137	in	4.45	4.60	4.525	4.525	+0.000	+0.00%	yes
put	325.00	0.137	in	5.50	5.70	5.600	5.600	+0.000	+0.00%	yes
put	330.00	0.137	in	6.80	6.90	6.850	6.850	+0.000	+0.00%	yes
put	335.00	0.137	in	8.25	8.35	8.300	8.300	+0.000	+0.00%	yes
put	340.00	0.137	in	10.00	10.15	10.075	10.075	+0.000	+0.00%	yes
put	345.00	0.137	in	12.15	12.35	12.250	12.250	+0.000	+0.00%	yes
put	350.00	0.137	out, Am.	14.55	14.85	14.700	14.761	+0.061	+0.42%	yes
put	355.00	0.137	out, Am.	17.15	17.50	17.325	17.456	+0.131	+0.76%	yes
put	360.00	0.137	out, Am.	20.25	20.75	20.500	20.508	+0.008	+0.04%	yes
put	365.00	0.137	out, Am.	23.60	24.50	24.050	23.856	-0.194	-0.81%	yes
put	370.00	0.137	out, Am.	27.20	28.20	27.700	27.597	-0.103	-0.37%	yes
put	375.00	0.137	out, Am.	30.95	32.30	31.625	31.601	-0.024	-0.08%	yes
put	380.00	0.137	out, Am.	35.35	36.80	36.075	35.846	-0.229	-0.64%	yes
put	385.00	0.137	out, Am.	39.75	41.35	40.550	40.271	-0.279	-0.69%	yes
put	390.00	0.137	out, Am.	44.35	46.05	45.200	44.932	-0.268	-0.59%	yes
put	395.00	0.137	out, Am.	49.15	50.90	50.025	49.727	-0.298	-0.60%	yes
call	295.00	0.214	out	55.40	56.20	55.800	55.392	-0.408	-0.73%	yes
call	300.00	0.214	out	50.95	51.50	51.225	50.916	-0.309	-0.60%	no
call	305.00	0.214	out	46.45	47.20	46.825	46.646	-0.179	-0.38%	yes
call	310.00	0.214	out	42.25	42.75	42.500	42.335	-0.165	-0.39%	yes
call	315.00	0.214	out	38.20	38.55	38.375	38.324	-0.051	-0.13%	yes
call	320.00	0.214	out	34.25	34.80	34.525	34.364	-0.161	-0.47%	yes

Type	Strike	T (y)	Sample	Bid	Ask	Mid	Model	Error	Rel.	Hit
call	325.00	0.214	out	30.25	31.10	30.675	30.628	-0.047	-0.15%	yes
call	330.00	0.214	out	27.00	27.35	27.175	27.017	-0.158	-0.58%	yes
call	335.00	0.214	out	23.60	23.95	23.775	23.707	-0.068	-0.29%	yes
call	340.00	0.214	out	20.40	20.60	20.500	20.496	-0.004	-0.02%	yes
call	345.00	0.214	out	17.45	17.75	17.600	17.611	+0.011	+0.06%	yes
call	350.00	0.214	in	14.85	15.05	14.950	14.950	+0.000	+0.00%	yes
call	355.00	0.214	in	12.45	12.60	12.525	12.525	+0.000	+0.00%	yes
call	360.00	0.214	in	10.30	10.60	10.450	10.450	+0.000	+0.00%	yes
call	365.00	0.214	in	8.35	8.60	8.475	8.475	+0.000	+0.00%	yes
call	370.00	0.214	in	6.85	7.00	6.925	6.925	+0.000	+0.00%	yes
call	375.00	0.214	in	5.45	5.65	5.550	5.550	+0.000	+0.00%	yes
call	380.00	0.214	in	4.35	4.50	4.425	4.425	+0.000	+0.00%	yes
call	385.00	0.214	in	3.40	3.50	3.450	3.450	+0.000	+0.00%	yes
call	390.00	0.214	in	2.70	2.79	2.745	2.745	+0.000	+0.00%	yes
call	395.00	0.214	in	1.97	2.17	2.070	2.070	+0.000	+0.00%	yes
put	295.00	0.214	in	2.43	2.62	2.525	2.525	+0.000	+0.00%	yes
put	300.00	0.214	in	2.97	3.05	3.010	3.010	+0.000	+0.00%	yes
put	305.00	0.214	in	3.60	3.80	3.700	3.700	+0.000	+0.00%	yes
put	310.00	0.214	in	4.25	4.45	4.350	4.350	+0.000	+0.00%	yes
put	315.00	0.214	in	5.25	5.35	5.300	5.300	+0.000	+0.00%	yes
put	320.00	0.214	in	6.25	6.35	6.300	6.300	+0.000	+0.00%	yes
put	325.00	0.214	in	7.45	7.60	7.525	7.525	+0.000	+0.00%	yes
put	330.00	0.214	in	8.80	8.95	8.875	8.875	+0.000	+0.00%	yes
put	335.00	0.214	in	10.45	10.60	10.525	10.525	+0.000	+0.00%	yes
put	340.00	0.214	in	12.20	12.35	12.275	12.275	+0.000	+0.00%	yes
put	345.00	0.214	in	14.25	14.45	14.350	14.350	+0.000	+0.00%	yes
put	350.00	0.214	out, Am.	16.50	16.80	16.650	16.930	+0.280	+1.68%	no
put	355.00	0.214	out, Am.	19.10	19.60	19.350	19.532	+0.182	+0.94%	yes
put	360.00	0.214	out, Am.	22.05	22.40	22.225	22.497	+0.272	+1.23%	no
put	365.00	0.214	out, Am.	25.20	26.15	25.675	25.589	-0.086	-0.34%	yes
put	370.00	0.214	out, Am.	28.65	29.65	29.150	29.112	-0.038	-0.13%	yes
put	375.00	0.214	out, Am.	32.20	33.45	32.825	32.842	+0.017	+0.05%	yes
put	380.00	0.214	out, Am.	36.15	37.70	36.925	36.834	-0.091	-0.25%	yes
put	385.00	0.214	out, Am.	40.30	42.00	41.150	41.018	-0.132	-0.32%	yes
put	390.00	0.214	out, Am.	44.55	45.90	45.225	45.469	+0.244	+0.54%	yes
put	395.00	0.214	out, Am.	49.00	51.10	50.050	50.019	-0.031	-0.06%	yes
call	300.00	0.310	out	53.95	55.20	54.575	54.106	-0.469	-0.86%	yes
call	310.00	0.310	out	45.85	46.60	46.225	46.020	-0.205	-0.44%	yes
call	320.00	0.310	out	38.40	39.10	38.750	38.434	-0.316	-0.81%	yes
call	330.00	0.310	out	31.25	31.75	31.500	31.398	-0.102	-0.32%	yes
call	340.00	0.310	out	25.15	25.50	25.325	25.162	-0.163	-0.64%	yes
call	350.00	0.310	in	19.45	19.75	19.600	19.600	+0.000	+0.00%	yes

Type	Strike	T (y)	Sample	Bid	Ask	Mid	Model	Error	Rel.	Hit
call	360.00	0.310	in	14.75	15.05	14.900	14.900	+0.000	+0.00%	yes
call	370.00	0.310	in	10.75	11.15	10.950	10.950	+0.000	+0.00%	yes
call	380.00	0.310	in	7.75	8.00	7.875	7.875	+0.000	+0.00%	yes
call	390.00	0.310	in	5.55	5.70	5.625	5.625	+0.000	+0.00%	yes
put	300.00	0.310	in	5.00	5.25	5.125	5.125	+0.000	+0.00%	yes
put	310.00	0.310	in	6.80	7.05	6.925	6.925	+0.000	+0.00%	yes
put	320.00	0.310	in	9.10	9.35	9.225	9.225	+0.000	+0.00%	yes
put	330.00	0.310	in	11.90	12.25	12.075	12.075	+0.000	+0.00%	yes
put	340.00	0.310	in	15.45	16.00	15.725	15.725	+0.000	+0.00%	yes
put	350.00	0.310	out, Am.	19.90	20.40	20.150	20.439	+0.289	+1.44%	no
put	360.00	0.310	out, Am.	25.05	25.70	25.375	25.806	+0.431	+1.70%	no
put	370.00	0.310	out, Am.	31.15	32.60	31.875	31.975	+0.100	+0.31%	yes
put	380.00	0.310	out, Am.	38.40	39.95	39.175	39.082	-0.093	-0.24%	yes
put	390.00	0.310	out, Am.	45.60	48.40	47.000	47.101	+0.101	+0.22%	yes
call	300.00	0.386	out	56.25	57.15	56.700	56.393	-0.307	-0.54%	yes
call	310.00	0.386	out	48.10	49.20	48.650	48.487	-0.163	-0.34%	yes
call	320.00	0.386	out	40.90	41.85	41.375	41.055	-0.320	-0.77%	yes
call	330.00	0.386	out	34.30	34.80	34.550	34.223	-0.327	-0.95%	no
call	340.00	0.386	out	28.00	28.25	28.125	28.066	-0.059	-0.21%	yes
call	350.00	0.386	in	22.35	22.65	22.500	22.500	+0.000	+0.00%	yes
call	360.00	0.386	in	17.60	17.85	17.725	17.725	+0.000	+0.00%	yes
call	370.00	0.386	in	13.50	13.70	13.600	13.600	+0.000	+0.00%	yes
call	380.00	0.386	in	10.20	10.35	10.275	10.275	+0.000	+0.00%	yes
call	390.00	0.386	in	7.55	7.70	7.625	7.625	+0.000	+0.00%	yes
put	300.00	0.386	in	6.40	6.55	6.475	6.475	+0.000	+0.00%	yes
put	310.00	0.386	in	8.30	8.55	8.425	8.425	+0.000	+0.00%	yes
put	320.00	0.386	in	10.75	10.95	10.850	10.850	+0.000	+0.00%	yes
put	330.00	0.386	in	13.75	14.00	13.875	13.875	+0.000	+0.00%	yes
put	340.00	0.386	in	17.50	17.65	17.575	17.575	+0.000	+0.00%	yes
put	350.00	0.386	out, Am.	21.75	22.10	21.925	22.371	+0.446	+2.04%	no
put	360.00	0.386	out, Am.	26.80	27.25	27.025	27.647	+0.622	+2.30%	no
put	370.00	0.386	out, Am.	32.35	33.30	32.825	33.635	+0.810	+2.47%	no
put	380.00	0.386	out, Am.	39.90	41.15	40.525	40.501	-0.024	-0.06%	yes
put	390.00	0.386	out, Am.	47.30	48.85	48.075	48.127	+0.052	+0.11%	yes
call	295.00	0.463	out	62.05	63.95	63.000	62.624	-0.376	-0.60%	yes
call	300.00	0.463	out	57.90	59.80	58.850	58.711	-0.139	-0.24%	yes
call	305.00	0.463	out	54.20	56.05	55.125	54.747	-0.378	-0.69%	yes
call	310.00	0.463	out	50.75	52.20	51.475	50.908	-0.567	-1.10%	yes
call	315.00	0.463	out	47.10	48.05	47.575	47.369	-0.206	-0.43%	yes
call	320.00	0.463	out	43.40	44.50	43.950	43.605	-0.345	-0.78%	yes
call	325.00	0.463	out	40.50	40.85	40.675	40.291	-0.384	-0.94%	no
call	330.00	0.463	out	36.90	37.50	37.200	37.003	-0.197	-0.53%	yes

Type	Strike	T (y)	Sample	Bid	Ask	Mid	Model	Error	Rel.	Hit
call	335.00	0.463	out	33.85	34.20	34.025	33.864	-0.161	-0.47%	yes
call	340.00	0.463	out	30.65	31.15	30.900	30.900	+0.000	+0.00%	yes
call	345.00	0.463	out	28.00	28.25	28.125	28.086	-0.039	-0.14%	yes
call	350.00	0.463	in	25.25	25.55	25.400	25.400	+0.000	+0.00%	yes
call	355.00	0.463	in	22.75	22.95	22.850	22.850	+0.000	+0.00%	yes
call	360.00	0.463	in	20.35	20.50	20.425	20.425	+0.000	+0.00%	yes
call	365.00	0.463	in	18.20	18.40	18.300	18.300	+0.000	+0.00%	yes
call	370.00	0.463	in	16.05	16.30	16.175	16.175	+0.000	+0.00%	yes
call	375.00	0.463	in	14.30	14.45	14.375	14.375	+0.000	+0.00%	yes
call	380.00	0.463	in	12.60	12.75	12.675	12.675	+0.000	+0.00%	yes
call	385.00	0.463	in	10.85	11.25	11.050	11.050	+0.000	+0.00%	yes
call	390.00	0.463	in	9.40	9.75	9.575	9.575	+0.000	+0.00%	yes
call	395.00	0.463	in	8.40	8.60	8.500	8.500	+0.000	+0.00%	yes
put	295.00	0.463	in	6.65	6.85	6.750	6.750	+0.000	+0.00%	yes
put	300.00	0.463	in	7.60	7.90	7.750	7.750	+0.000	+0.00%	yes
put	305.00	0.463	in	8.55	8.85	8.700	8.700	+0.000	+0.00%	yes
put	310.00	0.463	in	9.60	9.95	9.775	9.775	+0.000	+0.00%	yes
put	315.00	0.463	in	11.10	11.20	11.150	11.150	+0.000	+0.00%	yes
put	320.00	0.463	in	12.15	12.45	12.300	12.300	+0.000	+0.00%	yes
put	325.00	0.463	in	13.75	14.05	13.900	13.900	+0.000	+0.00%	yes
put	330.00	0.463	in	15.40	15.65	15.525	15.525	+0.000	+0.00%	yes
put	335.00	0.463	in	17.20	17.40	17.300	17.300	+0.000	+0.00%	yes
put	340.00	0.463	in	19.15	19.35	19.250	19.250	+0.000	+0.00%	yes
put	345.00	0.463	in	21.20	21.50	21.350	21.350	+0.000	+0.00%	yes
put	350.00	0.463	out, Am.	23.60	23.80	23.700	24.214	+0.514	+2.17%	no
put	355.00	0.463	out, Am.	25.90	26.30	26.100	26.685	+0.585	+2.24%	no
put	360.00	0.463	out, Am.	28.70	29.10	28.900	29.295	+0.395	+1.37%	no
put	365.00	0.463	out, Am.	31.20	31.75	31.475	32.217	+0.742	+2.36%	no
put	370.00	0.463	out, Am.	34.50	34.90	34.700	35.161	+0.461	+1.33%	no
put	375.00	0.463	out, Am.	37.40	38.60	38.000	38.442	+0.442	+1.16%	yes
put	380.00	0.463	out, Am.	40.90	42.05	41.475	41.846	+0.371	+0.90%	yes
put	385.00	0.463	out, Am.	43.85	45.65	44.750	45.356	+0.606	+1.35%	yes
put	390.00	0.463	out, Am.	48.00	49.95	48.975	49.045	+0.070	+0.14%	yes
put	395.00	0.463	out, Am.	51.45	54.05	52.750	53.121	+0.371	+0.70%	yes
call	300.00	0.633	out	63.50	64.25	63.875	63.390	-0.485	-0.76%	no
call	310.00	0.633	out	55.70	57.40	56.550	56.025	-0.525	-0.93%	yes
call	320.00	0.633	out	49.20	49.90	49.550	49.161	-0.389	-0.79%	no
call	330.00	0.633	out	42.40	43.35	42.875	42.772	-0.103	-0.24%	yes
call	340.00	0.633	out	36.40	37.20	36.800	36.782	-0.018	-0.05%	yes
call	350.00	0.633	in	31.00	31.65	31.325	31.325	+0.000	+0.00%	yes
call	360.00	0.633	in	26.15	26.60	26.375	26.375	+0.000	+0.00%	yes
call	370.00	0.633	in	21.65	22.15	21.900	21.900	+0.000	+0.00%	yes

Type	Strike	T (y)	Sample	Bid	Ask	Mid	Model	Error	Rel.	Hit
call	380.00	0.633	in	17.75	18.25	18.000	18.000	+0.000	+0.00%	yes
call	390.00	0.633	in	14.40	14.90	14.650	14.650	+0.000	+0.00%	yes
put	300.00	0.633	in	10.25	10.60	10.425	10.425	+0.000	+0.00%	yes
put	310.00	0.633	in	12.60	13.05	12.825	12.825	+0.000	+0.00%	yes
put	320.00	0.633	in	15.35	16.10	15.725	15.725	+0.000	+0.00%	yes
put	330.00	0.633	in	18.75	19.45	19.100	19.100	+0.000	+0.00%	yes
put	340.00	0.633	in	22.70	23.05	22.875	22.875	+0.000	+0.00%	yes
put	350.00	0.633	out, Am.	27.00	27.40	27.200	28.082	+0.882	+3.24%	no
put	360.00	0.633	out, Am.	31.80	32.40	32.100	33.151	+1.051	+3.28%	no
put	370.00	0.633	out, Am.	37.15	38.05	37.600	38.763	+1.163	+3.09%	no
put	380.00	0.633	out, Am.	43.50	45.10	44.300	45.026	+0.726	+1.64%	yes
put	390.00	0.633	out, Am.	50.65	52.15	51.400	51.917	+0.517	+1.01%	yes
call	295.00	0.885	out	72.95	74.95	73.950	73.497	-0.453	-0.61%	yes
call	300.00	0.885	out	69.45	70.20	69.825	69.761	-0.064	-0.09%	yes
call	305.00	0.885	out	66.15	67.60	66.875	66.325	-0.550	-0.82%	yes
call	310.00	0.885	out	62.50	64.15	63.325	62.864	-0.461	-0.73%	yes
call	315.00	0.885	out	58.95	60.30	59.625	59.504	-0.121	-0.20%	yes
call	320.00	0.885	out	55.90	57.25	56.575	56.218	-0.357	-0.63%	yes
call	325.00	0.885	out	52.60	53.85	53.225	53.207	-0.018	-0.03%	yes
call	330.00	0.885	out	49.45	50.90	50.175	50.071	-0.104	-0.21%	yes
call	335.00	0.885	out	46.55	47.90	47.225	47.161	-0.064	-0.14%	yes
call	340.00	0.885	out	43.80	44.70	44.250	44.250	+0.000	+0.00%	yes
call	345.00	0.885	out	41.30	42.00	41.650	41.564	-0.086	-0.21%	yes
call	350.00	0.885	in	38.25	39.20	38.725	38.725	+0.000	+0.00%	yes
call	355.00	0.885	in	36.05	36.60	36.325	36.325	+0.000	+0.00%	yes
call	360.00	0.885	in	33.75	34.25	34.000	34.000	+0.000	+0.00%	yes
call	365.00	0.885	in	30.90	31.80	31.350	31.350	+0.000	+0.00%	yes
call	370.00	0.885	in	28.80	29.55	29.175	29.175	+0.000	+0.00%	yes
call	375.00	0.885	in	26.85	27.40	27.125	27.125	+0.000	+0.00%	yes
call	380.00	0.885	in	24.85	25.35	25.100	25.100	+0.000	+0.00%	yes
call	385.00	0.885	in	23.20	23.45	23.325	23.325	+0.000	+0.00%	yes
call	390.00	0.885	in	21.20	21.65	21.425	21.425	+0.000	+0.00%	yes
call	395.00	0.885	in	19.50	20.05	19.775	19.775	+0.000	+0.00%	yes
put	295.00	0.885	in	12.05	12.90	12.475	12.475	+0.000	+0.00%	yes
put	300.00	0.885	in	13.35	13.80	13.575	13.575	+0.000	+0.00%	yes
put	305.00	0.885	in	14.60	15.35	14.975	14.975	+0.000	+0.00%	yes
put	310.00	0.885	in	16.10	16.60	16.350	16.350	+0.000	+0.00%	yes
put	315.00	0.885	in	17.60	18.05	17.825	17.825	+0.000	+0.00%	yes
put	320.00	0.885	in	19.05	19.70	19.375	19.375	+0.000	+0.00%	yes
put	325.00	0.885	in	20.75	21.65	21.200	21.200	+0.000	+0.00%	yes
put	330.00	0.885	in	22.60	23.20	22.900	22.900	+0.000	+0.00%	yes
put	335.00	0.885	in	24.45	25.20	24.825	24.825	+0.000	+0.00%	yes

Type	Strike	T (y)	Sample	Bid	Ask	Mid	Model	Error	Rel.	Hit
put	340.00	0.885	in	26.30	27.20	26.750	26.750	+0.000	+0.00%	yes
put	345.00	0.885	in	28.55	29.25	28.900	28.900	+0.000	+0.00%	yes
put	350.00	0.885	out, Am.	30.85	31.60	31.225	32.231	+1.006	+3.22%	no
put	355.00	0.885	out, Am.	33.20	34.20	33.700	34.812	+1.112	+3.30%	no
put	360.00	0.885	out, Am.	35.55	36.35	35.950	37.483	+1.533	+4.26%	no
put	365.00	0.885	out, Am.	38.25	39.40	38.825	39.872	+1.047	+2.70%	no
put	370.00	0.885	out, Am.	40.80	42.60	41.700	42.721	+1.021	+2.45%	no
put	375.00	0.885	out, Am.	43.90	45.55	44.725	45.731	+1.006	+2.25%	no
put	380.00	0.885	out, Am.	46.95	48.65	47.800	48.789	+0.989	+2.07%	no
put	385.00	0.885	out, Am.	49.90	52.10	51.000	52.084	+1.084	+2.13%	yes
put	390.00	0.885	out, Am.	53.80	55.25	54.525	55.330	+0.805	+1.48%	no
put	395.00	0.885	out, Am.	56.55	58.75	57.650	58.818	+1.168	+2.03%	no

Tests

The engine is covered by an automated pytest suite, run while generating this report. Offline tests use flat synthetic markets and a fake market data provider, so they are deterministic and need no network; the live comparison against Yahoo Finance is the previous section.

Command	pytest -m "not live"
Duration	40.0 s
Results	157 passed, 0 failed, 0 skipped, out of 157

Suite	What it checks	How	Tests	Pass	Fail
Closed-form Black-Scholes	Reference prices and analytic Greeks of the closed-form pricer.	Disabled: every test is commented out (written against the old snapshot API).	0	0	0
Binomial tree	European call and put prices of the Cox-Ross-Rubinstein tree, put-call parity.	Compares against known Black-Scholes values on a flat market (S=100, K=100, T=1).	3	3	0
Monte Carlo	Prices, put-call parity and price statistics for every variance reduction technique.	Tolerances in standard errors (4 SE) against the closed form; antithetic and control variate must lower the standard error.	20	20	0
Cross-model agreement	Monte Carlo and binomial tree agree with the closed form.	Same option and market priced by each pricer: tree within 0.001, Monte Carlo within 4 standard errors.	2	2	0
Numerical convergence	Errors shrink as the numerical resolution grows.	Tree with 10 / 50 / 1000 steps: error must decrease. MC with 1k / 50k / 200k paths: standard error must fall like 1 / sqrt(paths).	2	2	0
No-arbitrage bounds	Model-free price bounds for every pricer.	Checks e.g. max(S e^{-qT} - K e^{-rT}, 0) <= C <= S and the put equivalents.	24	24	0
American options	Early exercise on the binomial tree.	American put-call inequality, American >= European, call bounds with dividends.	5	5	0
Extreme regimes	Stability with extreme spot, rate, volatility and dividend inputs.	Parametrised stress scenarios: prices stay finite, within bounds, with the right limits (e.g. sigma -> 0).	16	16	0
Monte Carlo randomness	Reproducibility with a fixed seed and stability across runs.	Same seed gives identical prices; repeated runs differ, by less than 4 standard errors.	2	2	0
Input validation	Invalid inputs are rejected with clear errors.	Negative or non-integer steps / paths / seeds, wrong scheme types, invalid strikes, maturities, option types.	12	12	0
Market data module	Loaders, curves, implied volatility, vol surface, put-call parity calibration of spot and carry.	Offline: a fake provider serves a synthetic chain priced from a known smile, which the surface must recover.	52	52	0
Report generation	Market comparison, test result parsing and PDF rendering.	Offline: a fake market with a 1% spread must be repriced inside its spread; the PDF renders with and without optional sections.	19	19	0
Live market comparison	Model prices against real bid/ask quotes.	Hits Yahoo Finance; run with pytest -m live. Its results are the market comparison section of this report.	0	0	0

Limitations

Read the numbers above with these in mind.

- **Black-Scholes dynamics.** Each option gets a single volatility read from the surface. The model has no smile dynamics, stochastic volatility or jumps, so Greeks are sticky-strike and exotic or path-dependent products would be mispriced.

- **American options.** Only the binomial tree handles early exercise; the closed form and Monte Carlo are European only and the pipeline refuses them for American options. Implied volatilities are extracted with the European formula although listed options are American, which slightly overstates put volatilities.
- **Rates.** Treasury constant-maturity yields from local FRED files, used as continuous zero rates. The files must be refreshed manually; stale rates are partly absorbed by the implied carry.
- **Implied carry.** Put-call parity absorbs dividend and borrow errors exactly but rate errors only to first order. With the market closed (no bid/ask), the engine falls back to the quoted spot and the trailing dividend yield, and prices are less reliable.
- **Market data.** Yahoo Finance quotes are delayed and the equity and option feeds are not synchronised. Very short-dated, illiquid or far out-of-the-money quotes are excluded from the surface.
- **Volatility surface.** Natural cubic splines through raw implied vols with flat extrapolation in strike and maturity; no smoothing and no static-arbitrage checks.
- **Numerical error.** Tree prices converge with the step count and Monte Carlo prices carry sampling noise that falls like 1 / sqrt(paths). Finite-difference Greeks depend on the bump sizes.
- **Market comparison.** Only the liquid region is tested, at a single point in time; a hit means the price is inside the quoted spread, not that the model is right.